

EX NO:1	Butterworth IIR Filters
DATE:	1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

Code:

```

clc; clear;

Fs = 10000;
Fp = 1000;
Fst = 1500;

% Design Butterworth filter
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
[b, a] = butter(N, Wn);

% Frequency response
...

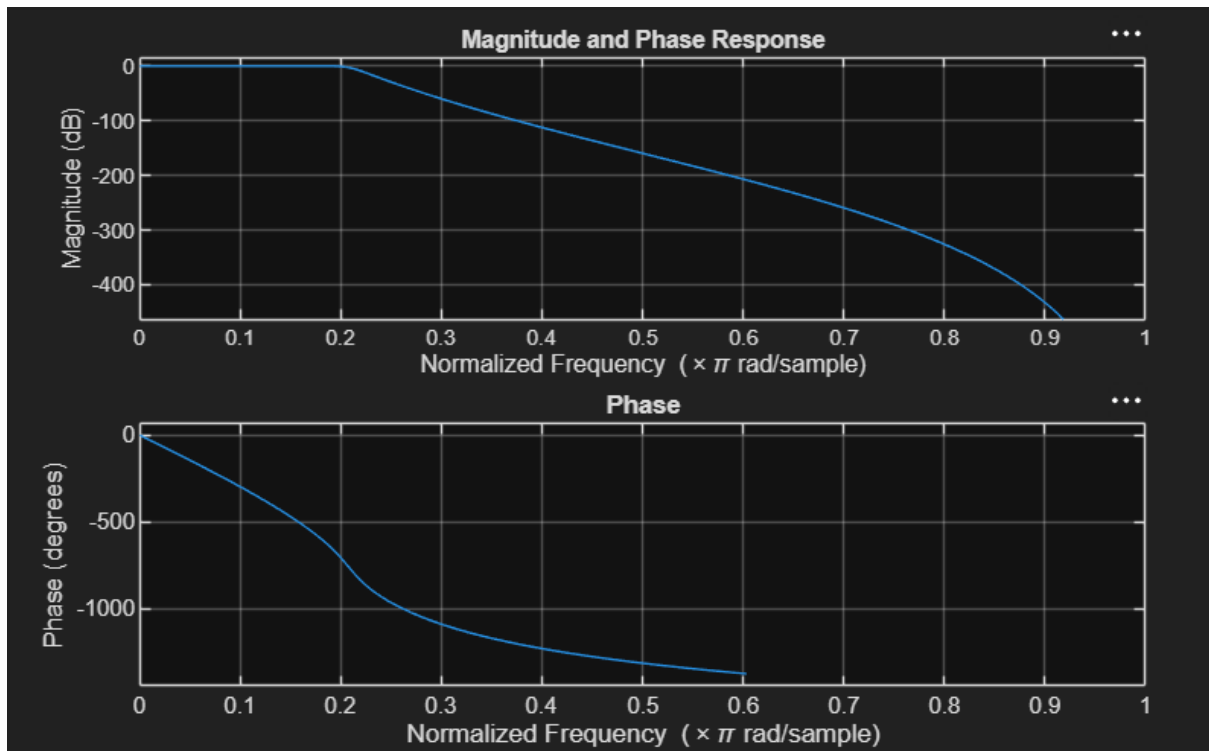
% Pole-zero plot
figure;
zplane(b, a);
title('Pole-Zero Plot');

% Display results
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);

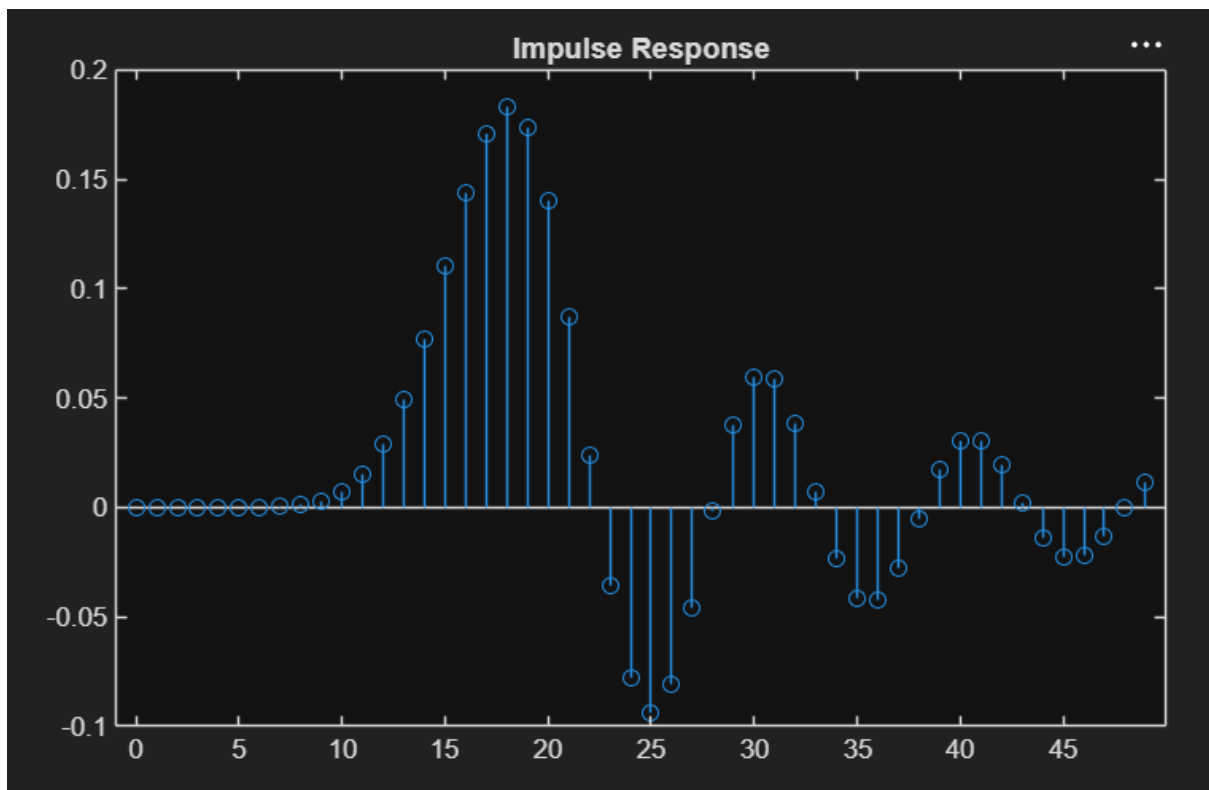
```

Output:

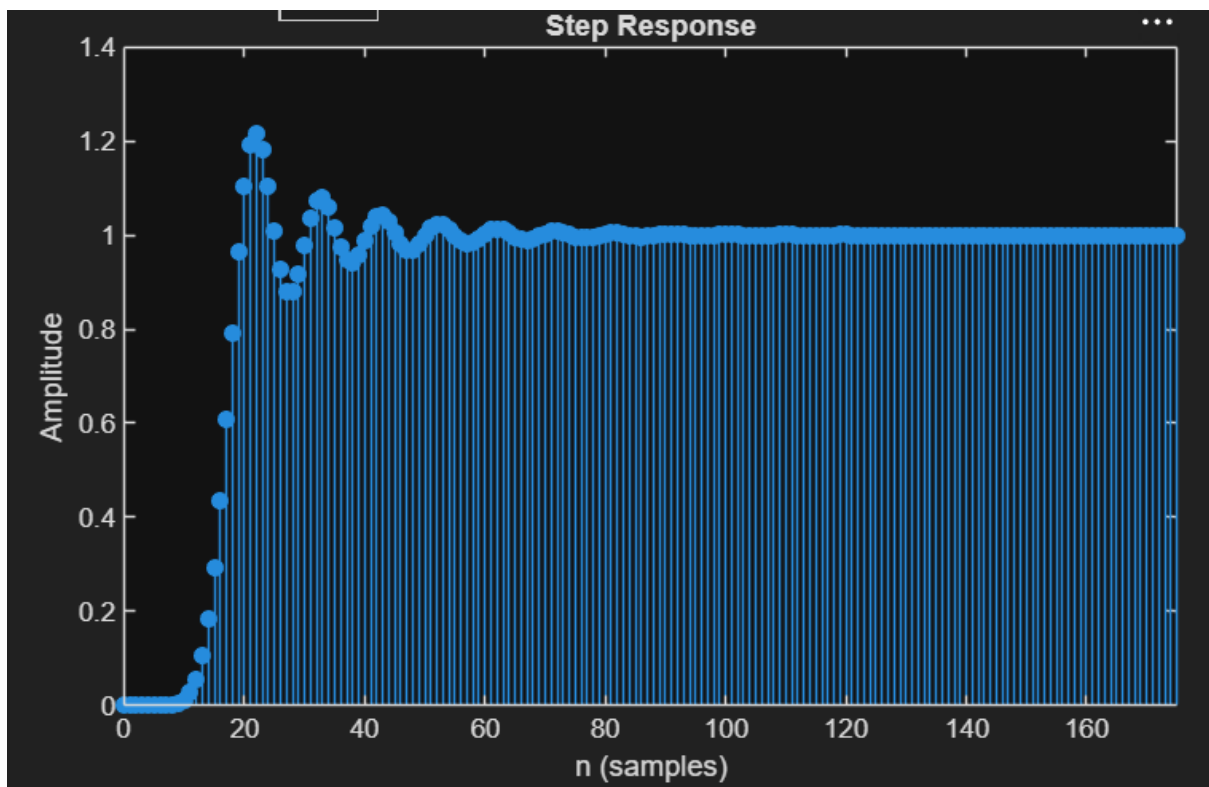
Objective 1:



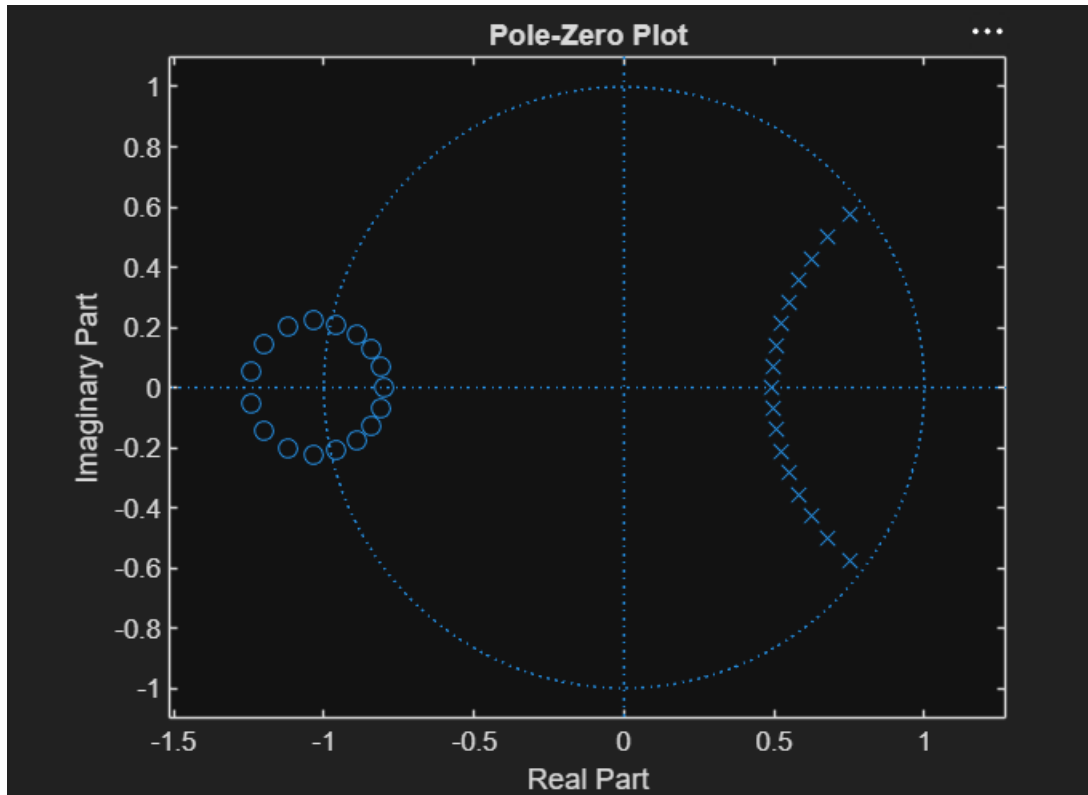
Objective 2:



Objective 3:



Objective 4:



EX NO:1	Butterworth IIR Filters
DATE:	1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

Code:

```
clc; clear;
```

```
% Specifications
```

```
Fs = 10000; % Sampling frequency
```

```
Fp = 3000; % Passband frequency
```

```
Fst = 2000; % Stopband frequency
```

```
% Filter design
```

```
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
```

```
[b, a] = butter(N, Wn, 'high');
```

```
...
```

```
% Pole-Zero Plot
```

```
figure;
```

```
zplane(b, a);
```

```
title('Pole-Zero Plot');
```

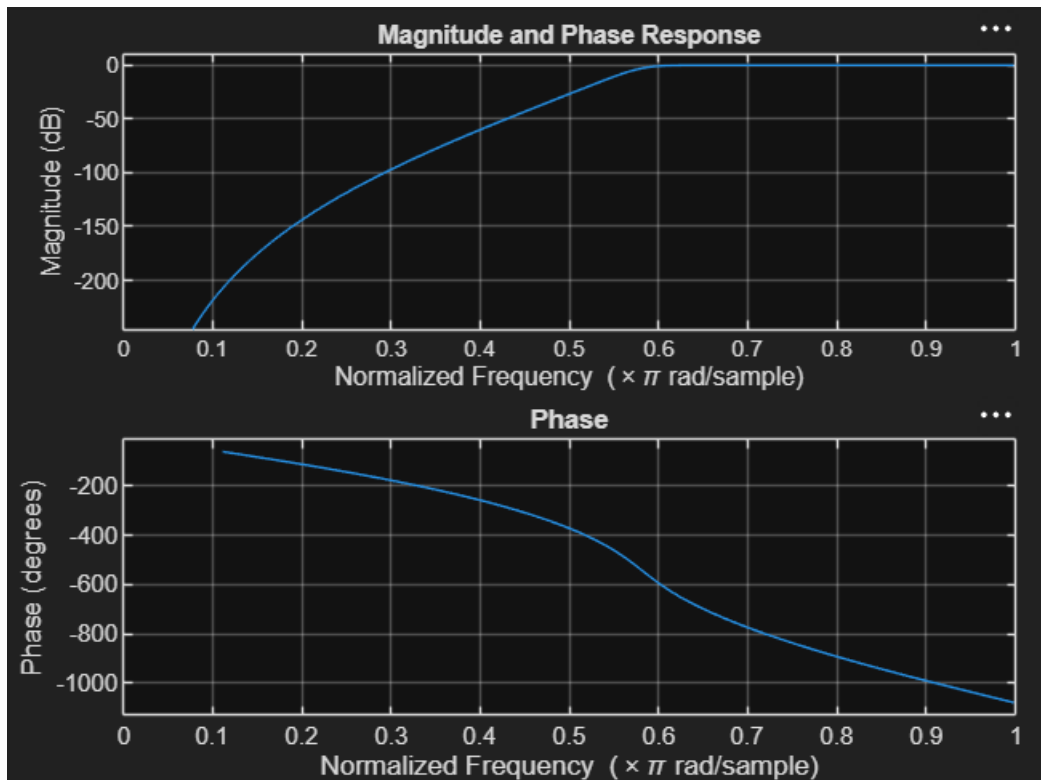
```
% Display results
```

```
fprintf('Minimum Filter Order (N) = %d\n', N);
```

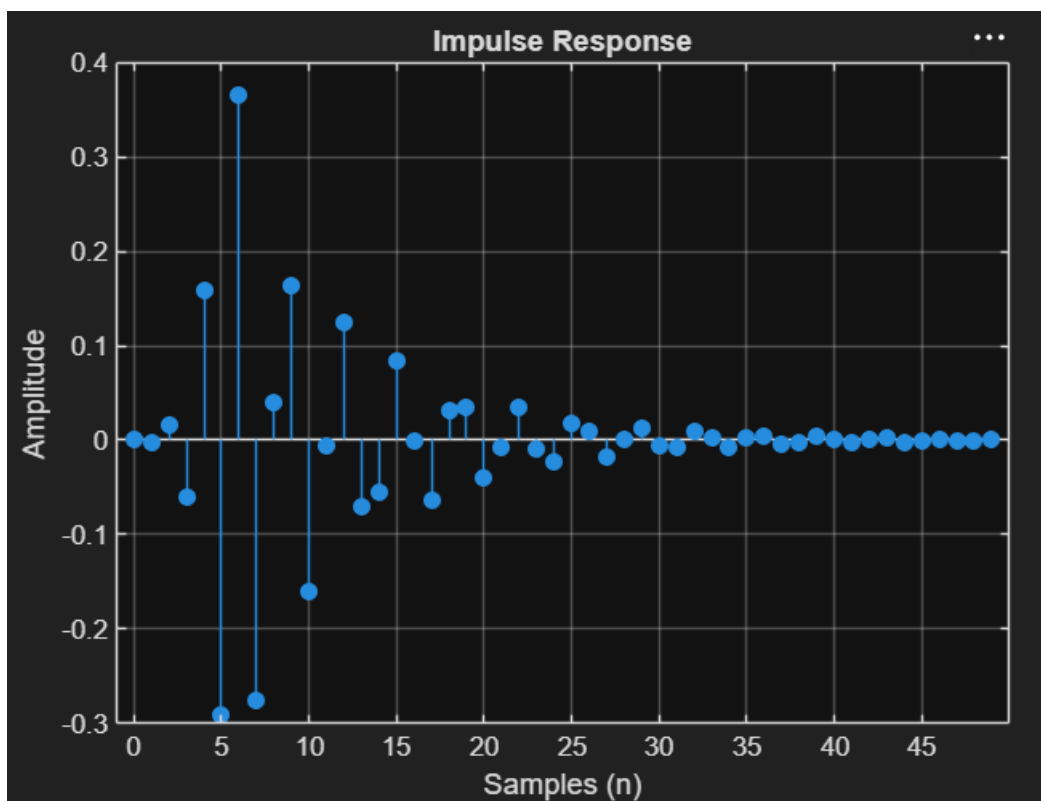
```
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```

Output:

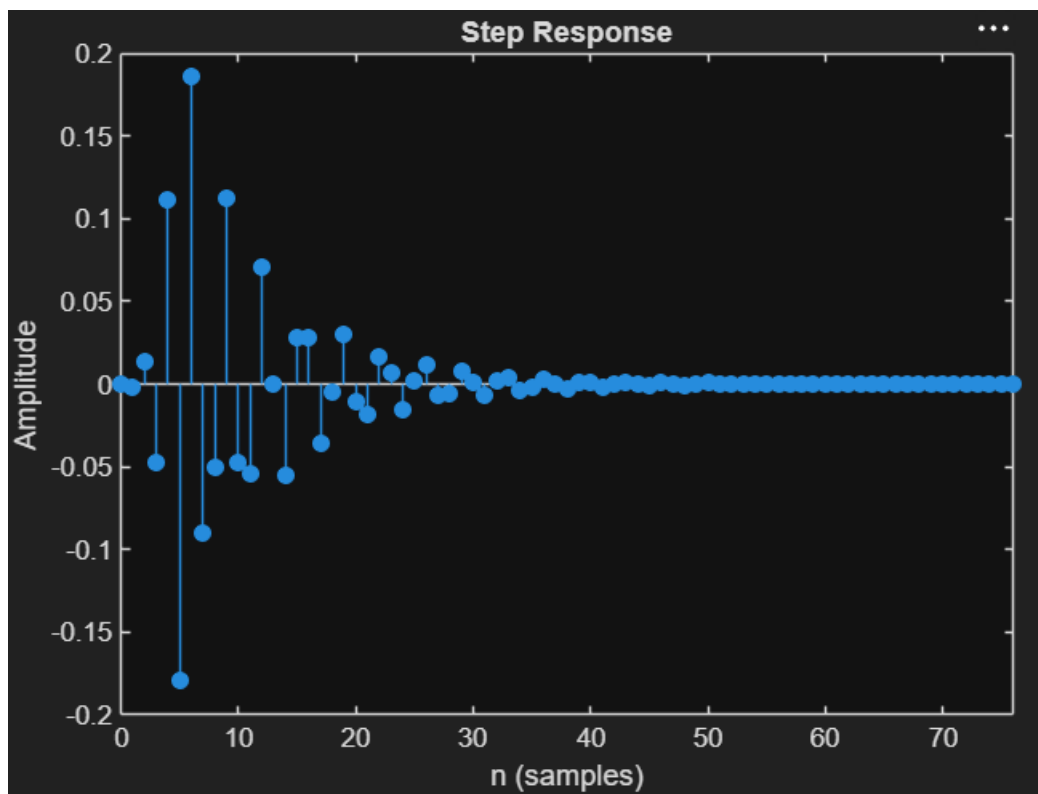
Objective 1:



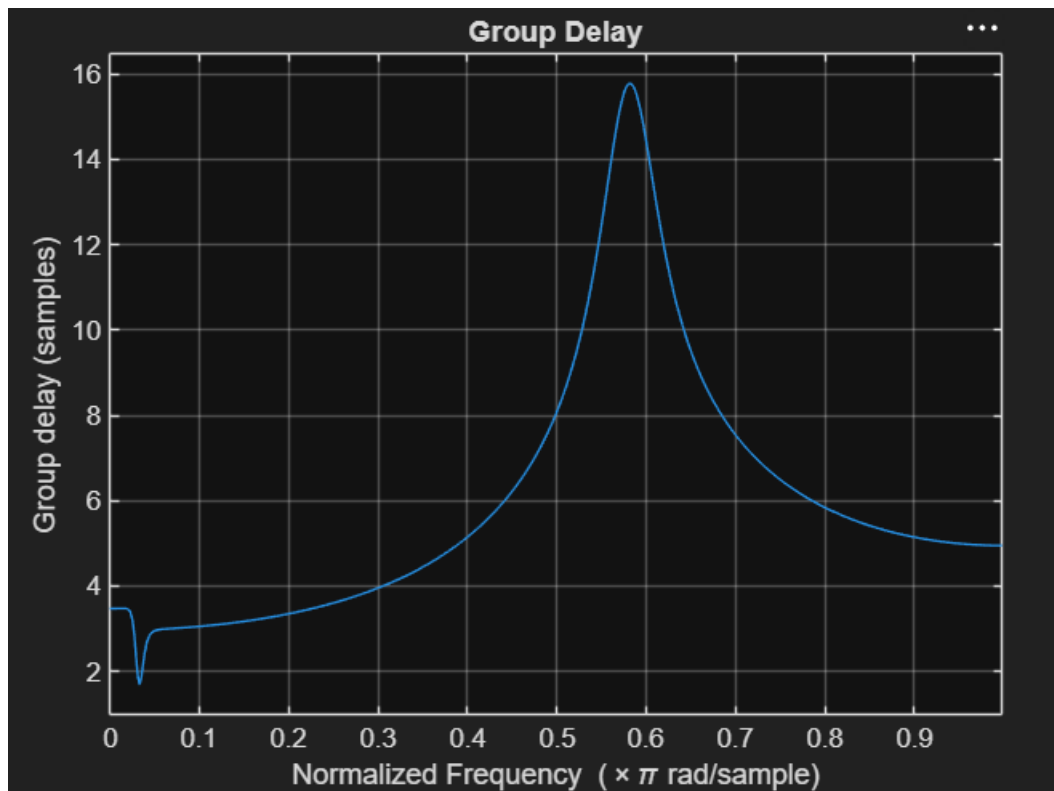
Objective 2:



Objective 3:



Objective 4:



Objective 5:

